

SKILL – 13:

PART 1 — INPUT REDIRECTION:

Code:

```
GNU nano 7.2 skill13_input.c *
#include <stdio.h>
#include <stdlib.h>

int main()
{
    FILE *file;
    file = fopen("input.txt", "r");
    if (file == NULL)
    {
        perror("Error opening input.txt");
        return 1;
    }
    if (freopen("input.txt", "r", stdin) == NULL)
    {
        perror("Error redirecting stdin");
        fclose(file);
        return 1;
    }
    printf("Input redirected successfully.\n");
    printf("Contents of input.txt:\n");
    char ch;
}
```

Output:

```
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-13$ ./skill13_input
Input redirected successfully.
Contents of input.txt:
Operating Systems
Input Redirection
File Handling in Linux
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-13$ |
```

Test missing file:

```
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-13$ mv input.txt input_backup.txt
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-13$ ./skill13_input
Error opening input.txt: No such file or directory
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-13$ |
```

PART 2 — OUTPUT REDIRECTION:

Code:

```
GNU nano 7.2 skill13_output.c *
#include <stdio.h>
#include <stdlib.h>

int main()
{
    FILE *file;
    file = fopen("output.txt", "w");
    if (file == NULL)
    {
        perror("Error creating output.txt");
        return 1;
    }
    if (freopen("output.txt", "w", stdout) == NULL)
    {
        perror("Error redirecting stdout");
        fclose(file);
        return 1;
    }
    printf("Output redirection successful.\n");
    printf("This message is written into output.txt.\n");
    printf("File handling and stdout redirection tested successfully.\n");
}
```

Output:

```
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-13$ ./skill13_output
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-13$ cat output.txt
Output redirection successful.
This message is written into output.txt.
File handling and stdout redirection tested successfully.
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-13$ |
```

Verify output file:

```
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-13$ ls -l output.txt
-rw-r--r-- 1 bhardwaj bhardwaj 130 Oct  7 15:38 output.txt
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-13$ cat output.txt
Output redirection successful.
This message is written into output.txt.
File handling and stdout redirection tested successfully.
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-13$ |
```

PART 3 — TEST NORMAL SHELL REDIRECTION:

Output:

```
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-13$ echo "Linux Output Redirection Test" > test_output.txt
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-13$ cat test_output.txt
Linux Output Redirection Test
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-13$ echo "Second line added using append redirection" >> test_output.txt
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-13$ cat test_output.txt
Linux Output Redirection Test
Second line added using append redirection
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-13$ |
```

PART 4 — TEST INPUT REDIRECTION USING <

```
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-13$ echo "Hello from redirected input" > test_input.txt
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-13$ cat < test_input.txt
Hello from redirected input
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-13$ |
```